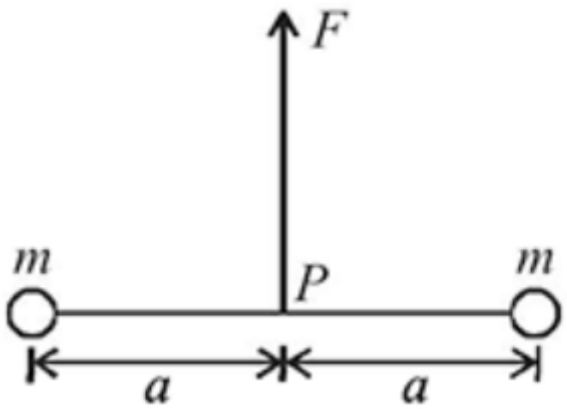
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two particles of mass m each are tied at the ends of a light string of length $2a$. The whole system is kept on a frictionless horizontal surface with the string held tight so that each mass is at a distance $'a'$ from the centre P (as shown in the figure). Now, the mid-point of the string is pulled vertically upwards with a small but constant force F. As a result, the particles move towards each other on the surface. The magnitude of acceleration, when the separation between them becomes $2x$, is



(A) $\frac{F}{2m}\frac{a}{\sqrt{a^2-x^2}}$

(B) $\frac{F}{2m}\frac{x}{\sqrt{a^2-x^2}}$

(C) $\frac{F}{2m}\frac{x}{a}$

(D) $\frac{F}{2m}\frac{\sqrt{a^2-x^2}}{x}$

Q.2

A boy, sitting on the topmost birth in the compartment of a train which is just going to stop on the railway station, drops an apple aiming at the open hand of his brother situated vertically below his own hand at a distance of $2\ \mathrm{m}$. The apple will fall

(A) in the hand of his brother

(B) slightly away from the hand of his brother in the direction of motion of the train

(C) slightly away from the hand of his brother opposite to the direction of motion of the train

(D) None of the above

Q.3

A block of mass m is resting on a smooth horizontal surface. One end of a uniform rope of mass $(\mathrm{m}/3)$ is fixed to the block, which is pulled in the horizontal direction by applying a force F at the other end. The tension in the middle of the rope is

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(A) $\frac{8}{7}$ F

(B) $\frac{1}{7}$ F

(C) $\frac{1}{8}$ F

(D) $\frac{7}{8}$ F

Q.4

An object of mass $10\ \text{kg}$ moves at a constant speed of $10\ \text{ms}^{-1}$. A constant force, that acts for 4 sec on the object, gives it a speed of $2\ \text{ms}^{-1}$ in opposite direction. The force acting on the object is

(A) $-3\ \text{N}$

(B) $-30\ \text{N}$

(C) $3\ \text{N}$

(D) $30\ \text{N}$

Q.5

If n bullets each of mass m are fired with a velocity v per second from a machine gun, the force required to hold the gun in position is

(A) $(n+1)mv$

(B) $\frac{mv}{n^2}$

(C) $\frac{mv}{n}$

(D) mnv

Q.6

A ball of mass $10\ \text{g}$ moving perpendicular to the plane of the wall strikes it and rebounds in the same line with the same velocity. If the impulse experienced by the wall is 0.54Ns, the velocity of the ball is

(A) $27\ \text{ms}^{-1}$

(B) $3.7\ \text{ms}^{-1}$

(C) $54\ \text{ms}^{-1}$

(D) $37\ \text{ms}^{-1}$

Q.7

Two bodies of masses $1\ \text{kg}$ and $2\ \text{kg}$ moving with same velocities are stopped by the same force. Then the ratio of their stopping distances is

(A) $1:2$

(B) $2:1$

(C) $\sqrt{2}:1$

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(D) $1:\sqrt{2}$

Q.8

A hammer weighing $3\ \mathrm{kg}$ strikes the head of a nail with a speed of $2\ \mathrm{ms}^{-1}$ drives it by $1\ \mathrm{cm}$ into the wall. The impulse imparted to the wall is

(A) $6\mathrm{Ns}$

(B) $3\mathrm{Ns}$

(C) $2\mathrm{Ns}$

(D) $12\mathrm{Ns}$

Q.9

The force required to stop a car of mass $800\ \mathrm{kg}$, moving at a speed of $20\ \mathrm{ms}^{-1}$ over a distance of $25\ \mathrm{m}$ in $2.5\mathrm{sec}$ is

(A) $1200\ \mathrm{N}$

(B) $6400\ \mathrm{N}$

(C) $1600\ \mathrm{N}$

(D) $1800\ \mathrm{N}$

Q.10

A string of length L and mass M are lying on a honzontal table. A force F is applied at one of its ends. Tension in the string at a distance x from the ends at which force is applied is

(A) Zero

(B) F

(C) $F(L-x)/L$

(D) $F(L-x)/M$

Q.11

A shopper pushes a shopping cart of a store with a constant force of $75\ \mathrm{N}$ [forward]. The shopping cart exerts a force of $75\ \mathrm{N}$ [backward] on the shopper

(A) only if the velocity of the cart is constant.

(B) only if there is no friction between the cart and the floor.

(C) only if the velocity of the cart is increasing.

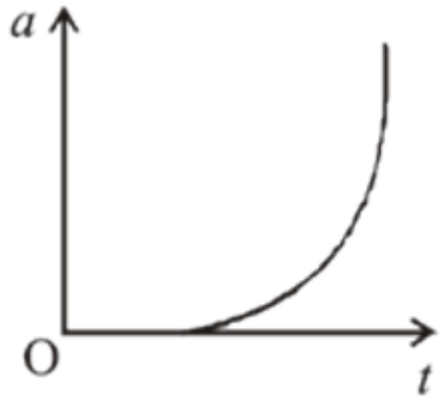
(D) under all circumstances assuming the system to be the shopper and cart.

Q.12

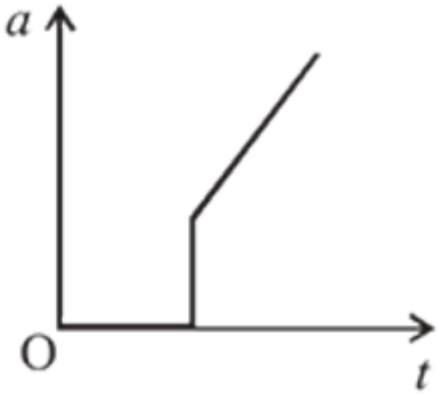
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A block is placed on a rough horizontal plane. A time dependent horizontal force $\mathrm{F} = \mathrm{kt}$ acts on the block, where k is a positive constant. The acceleration - time graph of the block is :

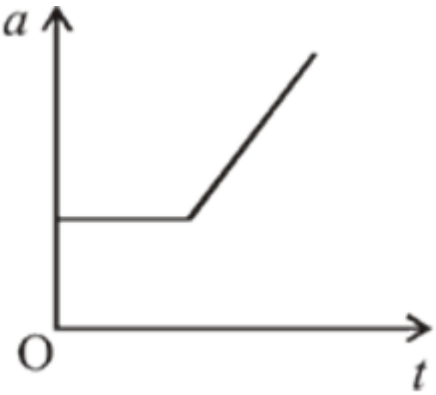
(A)



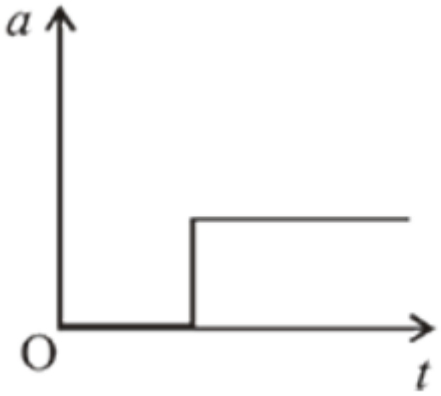
(B)



(C)



(D)



Q.13

A particle moves in the $X - Y$ plane under the influence of a force such that its linear momentum is $\vec{p}(t) = A[\hat{i}\cos(kt) - \hat{j}\sin(kt)]$, where A and k are constants. The angle between the force and the momentum is

(A) 0°

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(B) 30°

(C) 45°

(D) 90°

Q.14

A stone is dropped from a height h. It hits the ground with a certain momentum P. If the same stone is dropped from a height 100% more than the previous height, the momentum when it hits the ground will change by :

(A) 68%

(B) 41%

(C) 200%

(D) 100%

Q.15

Sand is being dropped on a conveyor belt at the rate of $\mathrm{Mkg/s}$. The force (in N) necessary to keep the belt moving with a constant velocity of $\mathrm{vm/s}$ will be:

(A) Mv

(B) $2\mathrm{Mv}$

(C) $\frac{\mathrm{Mv}}{2}$

(D) $\frac{\mathrm{Mv}}{3}$

Q.16

If a cricketer catches a ball of mass $150\mathrm{gm}$ moving with a velocity of $20\ \mathrm{m/s}$, then he experiences a force of (Time taken to complete the catch is $0.1\mathrm{sec.}$)

(A) $300\ \mathrm{N}$

(B) $30\ \mathrm{N}$

(C) $3\ \mathrm{N}$

(D) $0.3\ \mathrm{N}$

Q.17

The rate of mass of the gas emitted from rear of a rocket is initially $0.1\ \mathrm{kg/sec}$. If the speed of the gas relative to the rocket is $50\ \mathrm{m/sec}$ and mass of the rocket is $2\ \mathrm{kg}$, then the acceleration of the rocket in $\mathrm{m/sec^2}$ is

(A) 2.4

(B) 2.3

(C) 2.6

(D) 2.5

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Q.18

A ship of mass 3×10^7 kg initially at rest is pulled by a force of 5×10^4 N through a distance of 3 m. Assuming that the resistance due to water is negligible, the speed of the ship is

(A) 1.5 m/s

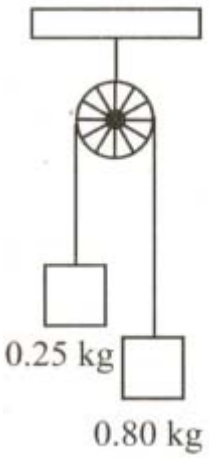
(B) 60 m/s

(C) 0.1 m/s

(D) 5 m/s

Q.19

Two blocks of masses 0.25 kg and 0.80 kg are connected by means of a light inextensible string that goes over a smooth and massless pulley. What is the work done (in joules) by string on the 0.25 kg block during the first second of its motion after being released from rest? (Take $g = 10$ m s^{-2})



(A) 09.95

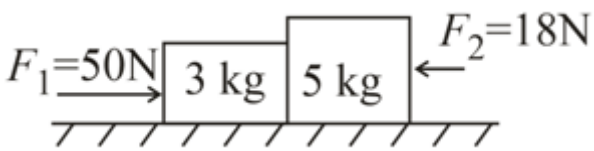
(B) 09.96

(C) 09.94

(D) 09.98

Q.20

Contact force between blocks is :-



(A) 50 N

(B) 38 N

(C) 32 N

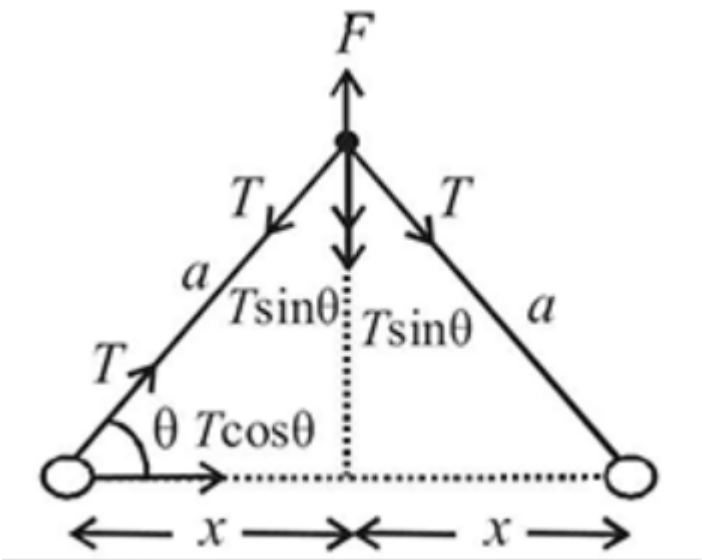
(D) 18 N

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Q.1 Answer:

$\frac{F}{2m}\frac{x}{\sqrt{a^2-x^2}}$

Solution:



The acceleration of mass m is due to the force $T\cos\theta$ $\therefore$ $T\cos\theta = ma \Rightarrow a = \frac{T\cos\theta}{m}$**...(i)**

also, $F = 2T\sin\theta \Rightarrow T = \frac{F}{2\sin\theta}$**...(ii)**

From (i) and (ii)

$$a = \left(\frac{F}{2\sin\theta}\right)\frac{\cos\theta}{F}$$

$$= \frac{F}{2m\tan\theta} = \frac{F}{2m}\frac{x}{\sqrt{a^2-x^2}}$$

$\left[\because \tan\theta = \frac{\sqrt{a^2-x^2}}{x}\right]$

Q.2 Answer:

slightly away from the hand of his brother in the direction of motion of the train

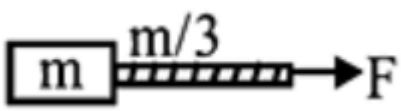
Solution:

The apple will fall slightly away from the hand of his brother in the direction of motion of the train due to inertia of motion. When train is just going to stop, the boy and his brother slows down with train but the apple which is in free fall continue to move with the same speed and therefore, falls slightly away from the hand in the direction of motion of the

Q.3 Answer:

$\frac{7}{8}$ F

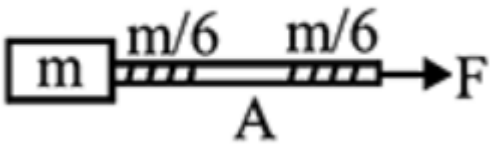
Solution:



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$\mathrm{m}_{\mathrm{m}^{\mathrm{m}}/3} \rightarrow \mathrm{F}$

The acceleration of the system is $\mathrm{a}=\frac{\mathrm{F}}{\mathrm{m}+\frac{\mathrm{m}}{3}}=\frac{3\,\mathrm{F}}{4\,\mathrm{m}} \quad \mathrm{m}\frac{\mathrm{m}/6\;\mathrm{m}/6}{\mathrm{A}}$



The tension in the middle of the rope (i.e., at point A **) is** $\mathrm{T}=\left(\mathrm{m}+\frac{\mathrm{m}}{6}\right)\mathrm{a}=\frac{7\,\mathrm{m}}{6}\times\frac{3\,\mathrm{F}}{4\,\mathrm{m}}\Rightarrow\mathrm{T}=\frac{7\,\mathrm{F}}{8}$

Q.4 Answer:

$-30\ \mathrm{N}$

Solution:

Here $u=10\ \mathrm{ms}^{-1}, \mathrm{v}=-2\ \mathrm{ms}^{-1},$
$\mathrm{t}=4\ \mathrm{s}, \mathrm{a}=?$
Using $\mathrm{a}=\frac{\mathrm{v}-\mathrm{u}}{\mathrm{t}}=\frac{-2-10}{4}=-3\ \mathrm{m/s}^2$
$\therefore$ **Force,** $\mathrm{F}=\mathrm{ma}=10\times(-3)=-30\ \mathrm{N}$

Q.5 Answer:

mnv

Solution:

By Newton's second law of motion

$$F=n(mv)=nmv$$

Q.6 Answer:

$27\ \mathrm{ms}^{-1}$

Solution:

As the ball, $\mathrm{m}=10\ \mathrm{g}=0.01\ \mathrm{kg}$ **rebounds after striking the wall** $\therefore$ **Change in momentum** $=mv-(-mv)=2mv$
Inpulse = Change in momentum $=2mv\ \therefore v=\frac{\text{Impulse}}{2m}=\frac{0.54\ \mathrm{N\,s}}{2\times0.01\ \mathrm{kg}}=27\ \mathrm{m\ s}^{-1}$

Q.7 Answer:

$1:2$

Solution:

$\mathrm{S}=\frac{\mathrm{V}^2-\mathrm{U}^2}{2\mathrm{a}}$ **and** $\mathrm{a}=\mathrm{F}/\mathrm{m}$**, so**
$\frac{S_1}{S_2}=\frac{\left(\frac{0-U^2}{2F/m}\right)}{\left(\frac{0-U^2}{2F/2m}\right)}=1:2$

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Q.8 Answer:

6Ns

Solution:

As we know, |impulse =| change in momentum $= |p_2 - p_1| = |0 - mv_1| = |0 - 3 \times 2| = 6Ns$

Q.9 Answer:

6400 N

Solution:

As we know, $|a| = \left|\frac{v^2 - u^2}{2S}\right| = \frac{400}{50} = 8\ m/s^2$

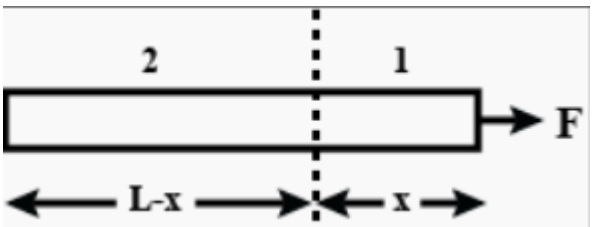
So force required $= F = ma$

$= 800 \times 8 = 6400\ N$

Q.10 Answer:

$F(L - x)/L$

Solution:



When force is applied, acceleration of constituting parts of the string $= a = F/M$

Mass of section $2 = m_2 = \frac{M}{L}(L - x)$

Applying Newton's second law on section 2 of the string,

$T = m_2 a$

$T = m_2 a = \frac{M}{L}(L - x)a$

$T = \frac{M}{L}(L - x)\frac{F}{M} = \frac{F}{L}(L - x)$

Q.11 Answer:

under all circumstances assuming the system to be the shopper and cart.

Solution:

This is an example of Newton's Third Law of Motion applicable in all circumstances

Q.12 Answer:



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Solution:

Graph (2) correctly depicts the accelerationtime graph of the block.

Q.13 Answer:

90°

Solution:

$\vec{p}(t) = A[\hat{i}\cos(kt) - \hat{j}\sin(kt)]$
$\vec{F} = \frac{d\vec{p}}{dt} = Ak[-\hat{i}\sin(kt) - \hat{j}\cos(kt)]$
Here, $\vec{F} \cdot \vec{P} = 0$ **But** $\vec{F} \cdot \vec{p} = Fp\cos\theta$
$\therefore \quad \cos\theta = 0 \Rightarrow \quad \theta = 90^{\circ}.$

Q.14 Answer:

41%

Solution:

Momentum $P = mv = m\sqrt{2gh}$
$(\because v^2 = u^2 + 2gh;$ **Here** $u = 0)$
When stone hits the ground momentum $P = m\sqrt{2gh}$
when same stone dropped from $2h(100\%$ **of initial) then momentum** $P' = m\sqrt{2g(2h)} = \sqrt{2}P$
Which is changed by 41% **of initial.**

Q.15 Answer:

Mv

Solution:

$F = \frac{d(Mv)}{dt} = M\frac{dv}{dt} + v\frac{dM}{dt}$
$\therefore$ **v is constant,** $\therefore F = v\frac{dM}{dt}$ **But** $\frac{dM}{dt} = M$ kg/s
$\therefore$ **F =** vM **newton.**

Q.16 Answer:

30 N

Solution:

Net force experienced $= \frac{\text{Total Impulse}}{\text{Time taken}} = \frac{m\Delta v}{t} = 0.15 \times \frac{20}{0.1} = 30$ N

Q.17 Answer:

2.5

Solution:

$\frac{\mathrm{dM}}{\mathrm{dt}} = 0.1$ kg/s, $\mathrm{v_{gas}} = 50$ m/s
Mass of the rocket $= 2$ kg.$\mathrm{Mv} =$ constant
$-\mathrm{v}\frac{\mathrm{dM}}{\mathrm{dt}} + \mathrm{M}\frac{\mathrm{dv}}{\mathrm{dt}} = 0 \quad \therefore \frac{\mathrm{dv}}{\mathrm{dt}} = \frac{1}{\mathrm{M}}\mathrm{v}\frac{\mathrm{dM}}{\mathrm{dt}}$

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$\Rightarrow$ **Acceleration** $= \frac{1}{2} \times 50 \times 0.1 = 2.5\ \mathrm{m/s^2}$

Q.18 Answer:

$0.1\ \mathrm{m/s}$

Solution:

$\mathrm{a} = \frac{\mathrm{F}}{\mathrm{m}} = \frac{5\times10^4}{3\times10^7} = \frac{5}{3} \times 10^{-3}\ \mathrm{m/s^2}$

velocity, $v = \sqrt{2\mathrm{as}}$

$= \sqrt{2 \times \left(\frac{5}{3} \times 10^{-3}\right) \times 3} = 0.1\ \mathrm{ms^{-1}}$

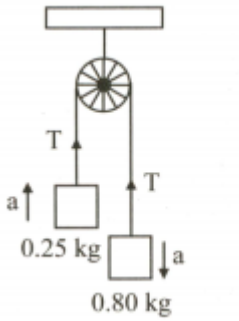
Q.19 Answer:

09.98

Solution:

Acceleration is given by, $\mathrm{a} = \frac{\text{Net pulling force}}{\text{Total mass}} = \frac{0.80\mathrm{g} - 0.25\mathrm{g}}{0.80 + 0.25} = \frac{11\mathrm{g}}{21}$

Displacement is given by, $\mathrm{s} = \frac{1}{2}\mathrm{at}^2 = \frac{1}{2}\frac{11\mathrm{g}}{21}(1)^2 = \frac{11\mathrm{g}}{42}$



Balancing force, we get,

$T - 0.25\ \ g = 0.25\ \ a$

$\therefore\ \ \mathrm{T} = \frac{1}{4}g + \frac{1}{4}\frac{11\mathrm{g}}{21}$

$\therefore\ \ \mathrm{T} = \frac{8}{21}\mathrm{g}$

Now, for $0.25\ \ \mathrm{kg}$ **mass,**

$\mathrm{W_T} = \mathrm{T}\cos0° = \frac{8}{21}\mathrm{g} \times \frac{11}{42}\mathrm{g} \times \mathrm{I} = \frac{44}{441}\mathrm{g}^2$

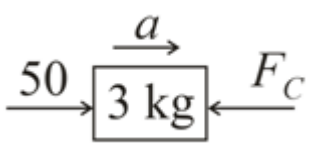
$\therefore\ \ W_T = 9.98\mathrm{J}$

Q.20 Answer:

$38\ \ \mathrm{N}$

Solution:

$a = \frac{50-18}{3+5} = \frac{32}{8} = 4\ \ \mathrm{m}\ \ \mathrm{s^{-2}}$



$50 - F_C = 3a$

$F_C = 50 - 3 \times 4$

$F_C = 38\ \ \mathrm{N}$

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